Introduction to Database System

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N-IT 2025 Informatics Study Program

Outline

- Database Characteristics
- DBMS features
- Architectures
- Organizational roles

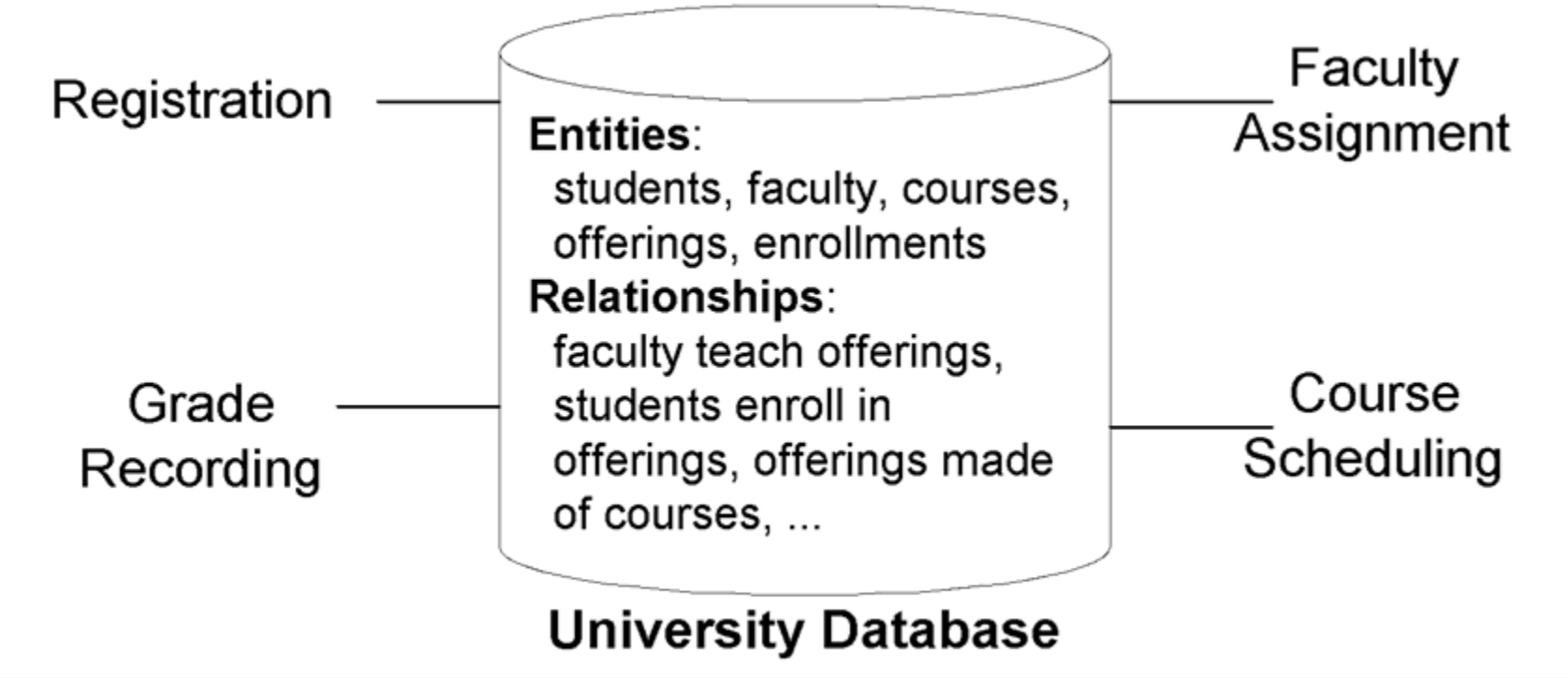
Initial Vocabulary

- Data: **raw facts** about things and events
- Information: **transformed data** that has value for decision making
- Essential to organize data for retrieval and maintenance

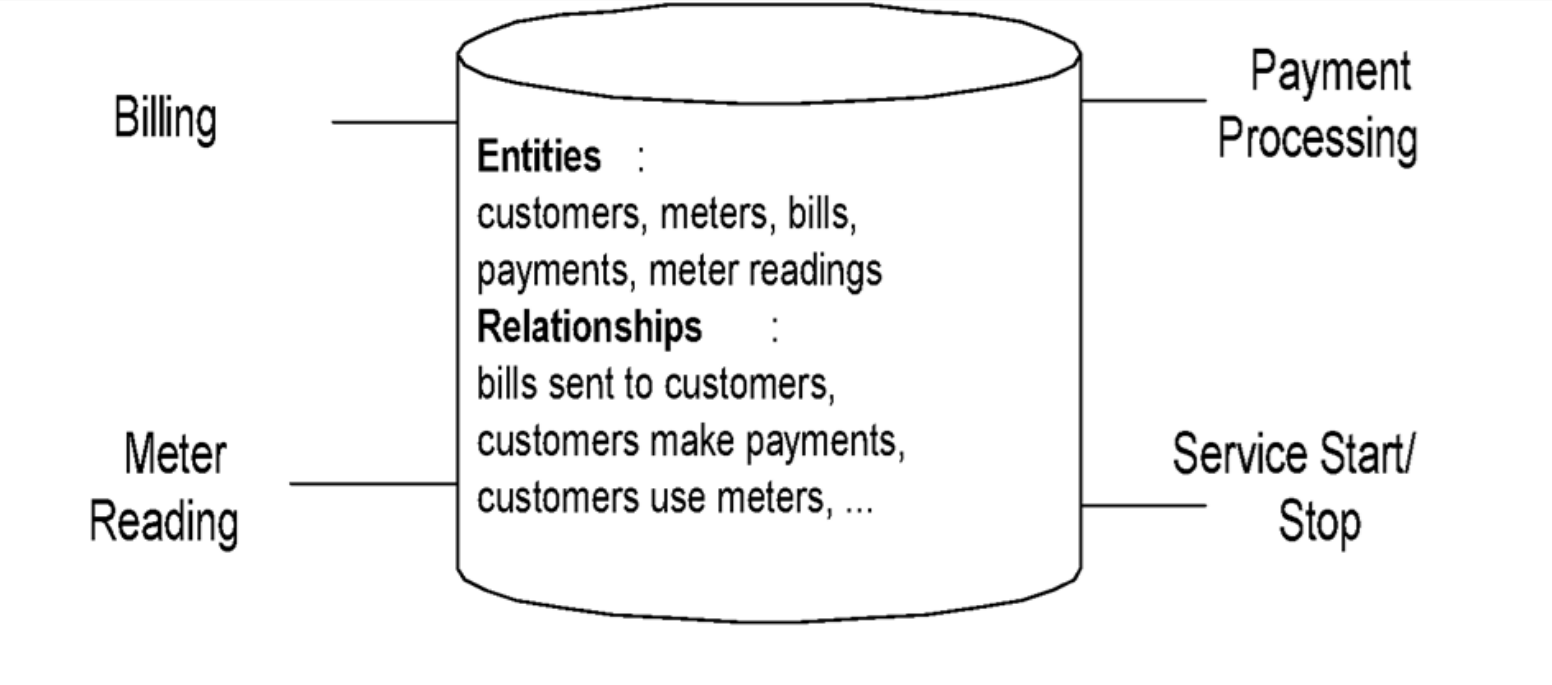
Database Characteristics

- Persistent : long time, potentially relevant data
- Inter related : entity-relationship
- Shared : simultaneously used by multiple user

Example of Database



Example of Database

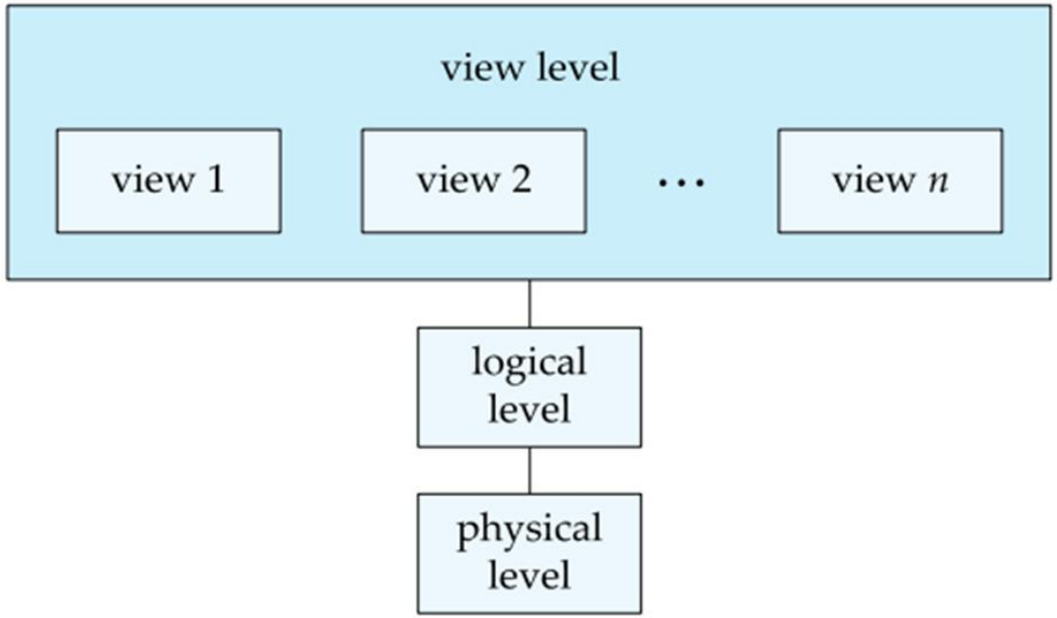


Purpose of Database System

- Data redundancy and inconsistency
- Difficulty in accessing data
- Data isolation
- Integrity problems
- Atomicity of updates
- Concurrent-access anomalies
- Security problems

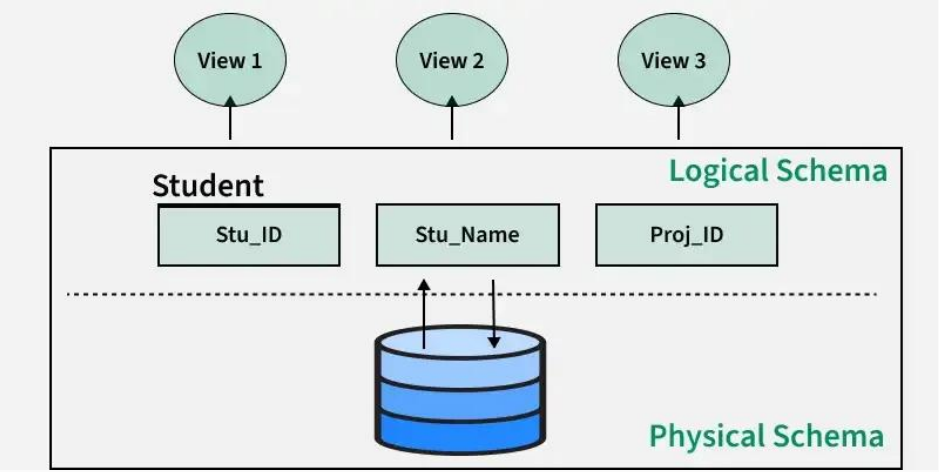
Database Systems offer solutions to all above problems

Architecture of Database System



Instances and Schemas

- Logical Schema
- Physical Schema
- Instances



Orders

order_id	item	amount	customer_id
1	Keyboard	400	4
2	Mouse	300	4
3	Monitor	12000	3
4	Keyboard	400	1
5	Mousepad	250	2

Schema = *what structure looks like*

Instance = *what data currently exists*

Physical Data Independence

- Ability to modify physical schema without changing the logical schema
- Flexibility, efficiency, easy maintenance
- Ex : server migration, data medium storage,

Logical Data Independence

- Ability to change logical schema without affecting the view schema or application programs
- Flexibility, LTS, compatibility with apps program
- Ex : add new columns, modify attribute

Database Access from Application Program

- SQL doesn't support action such as input from user, output to display, or communication over the network
- Computation and actions must be written in a host language (C/C++, Java, Python etc with embedded SQL queries that access the data in database
- Application programs - are programs that are used to interact with the database

DBMS

- Collection of components that support data acquisition, dissemination, storage, maintenance, retrieval, and formatting
- Enterprise DBMS : Oracle, Ms SQL Server, PostgreSQL, etc
- Desktop DBMS : Ms Access, Foxpro, etc
- Embedded DBMS : SQLite, Firebird, etc

Database Definition

- Define database before using
- Table and relationships
- SQL CREATE TABLE statement
- Graphical tools

Nonprocedural Access

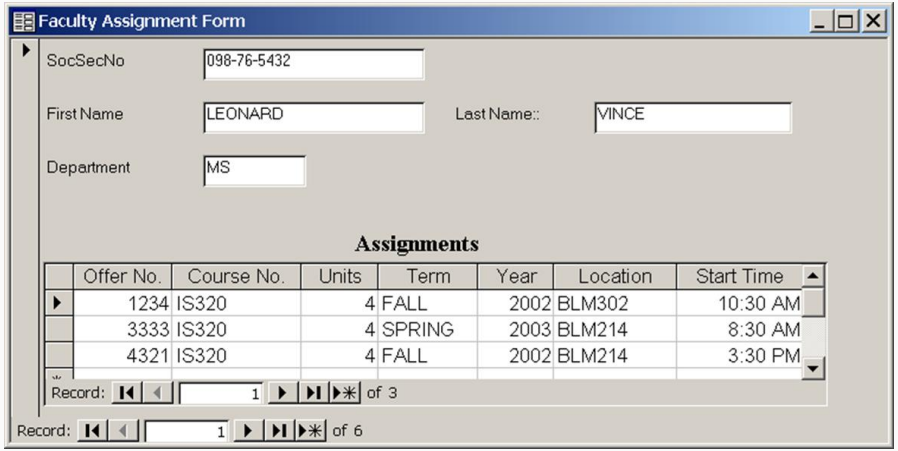
- Non procedural access : by state What not How
- Query: request for data to answer a question
- Indicate what parts of database to retrieve not the procedural details
- Improve productivity and improve accessibility
- SQL SELECT statement and graphical tools

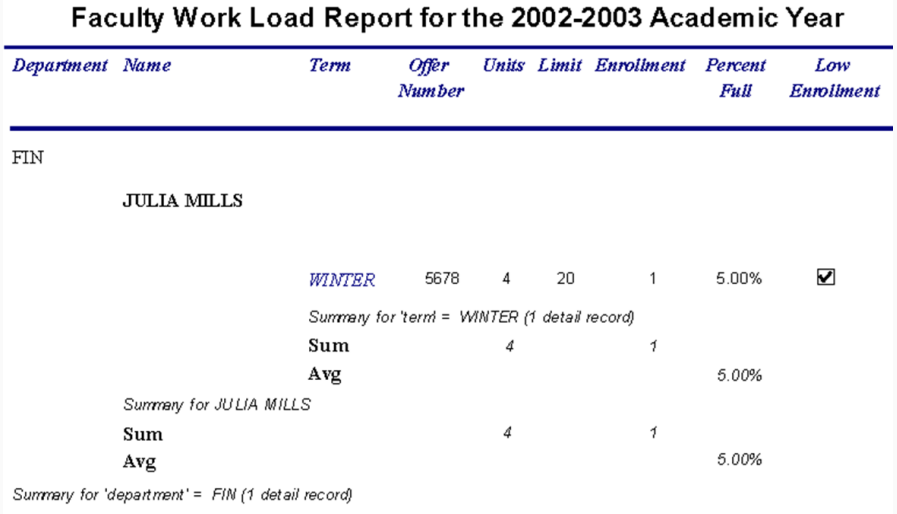
```
SELECT * FROM mahasiswa WHERE jurusan = 'Informatika';
```

Application Development

- Form: formatted document for data entry and display
- Report: formatted document for display
- Use nonprocedural access to specify data requirements of forms and reports

Application Development





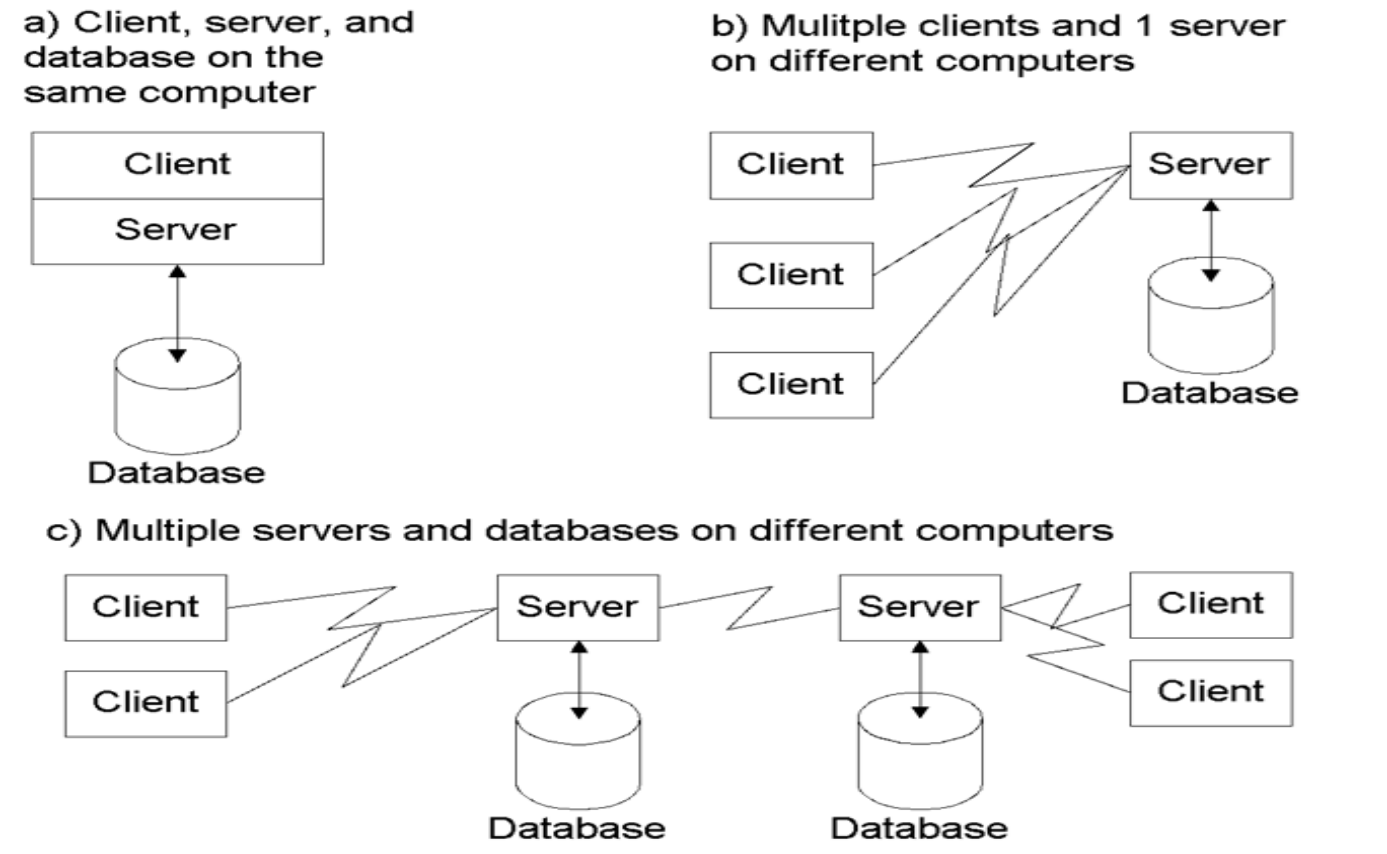
Faculty Work Load Report for the 2002-2003 Academic Year

Department	Name	Term	Offer Number	Units	Limit	Enrollment	Percent Full	Low Enrollment
FIN								
	JULIA MILLS							
		WINTER	5678	4	20	1	5.00%	☑
		Summary for 'term' = WINTER (1 detail record)						
		Sum		4		1		
		Avg					5.00%	
	Summary for JULIA MILLS							
	Sum			4		1		
	Avg						5.00%	
Summary for 'department' = FIN (1 detail record)								

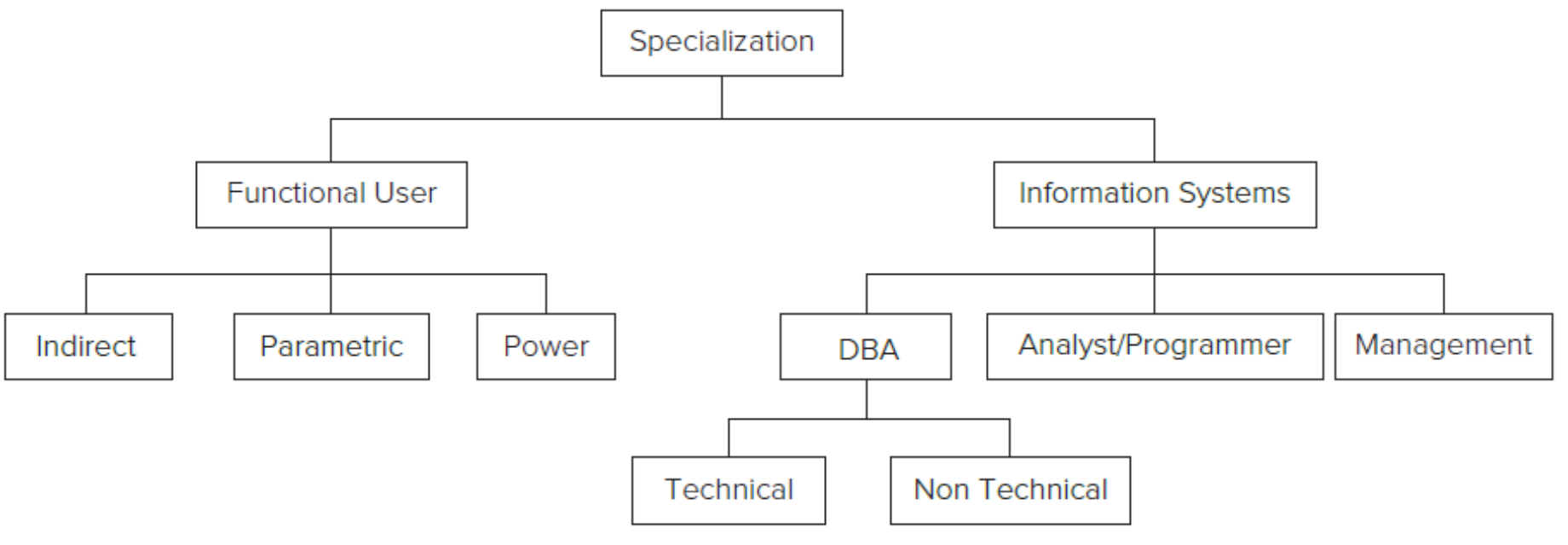
Procedural Language Interface

- Combine procedural language with nonprocedural access
- Why?
  - Batch processing
  - Customization and automation
  - Performance improvement

Client-Server Architecture



Organizational Roles



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The Relational Data Model

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Outline

- Relational model basics
- Integrity rules
- Rules about referenced rows
- Relational Algebra

Tables

- Relational database is a collection of tables
- Heading: table name and column names
- Body: rows, occurrences of data

StdSSN	StdLastName	StdMajor	StdClass	StdGPA
123-45-6789	WELLS	IS	FR	3.00
124-56-7890	NORBERT	FIN	JR	2.70
234-56-7890	KENDALL	ACCT	JR	3.50

Create Table Statement

- Define table name, column names, and column data types
- Convention:
  - Table names begin with uppercase
  - Mixed case for column names
  - First part of column name is an abbreviation for the table name
  - Upper case for data

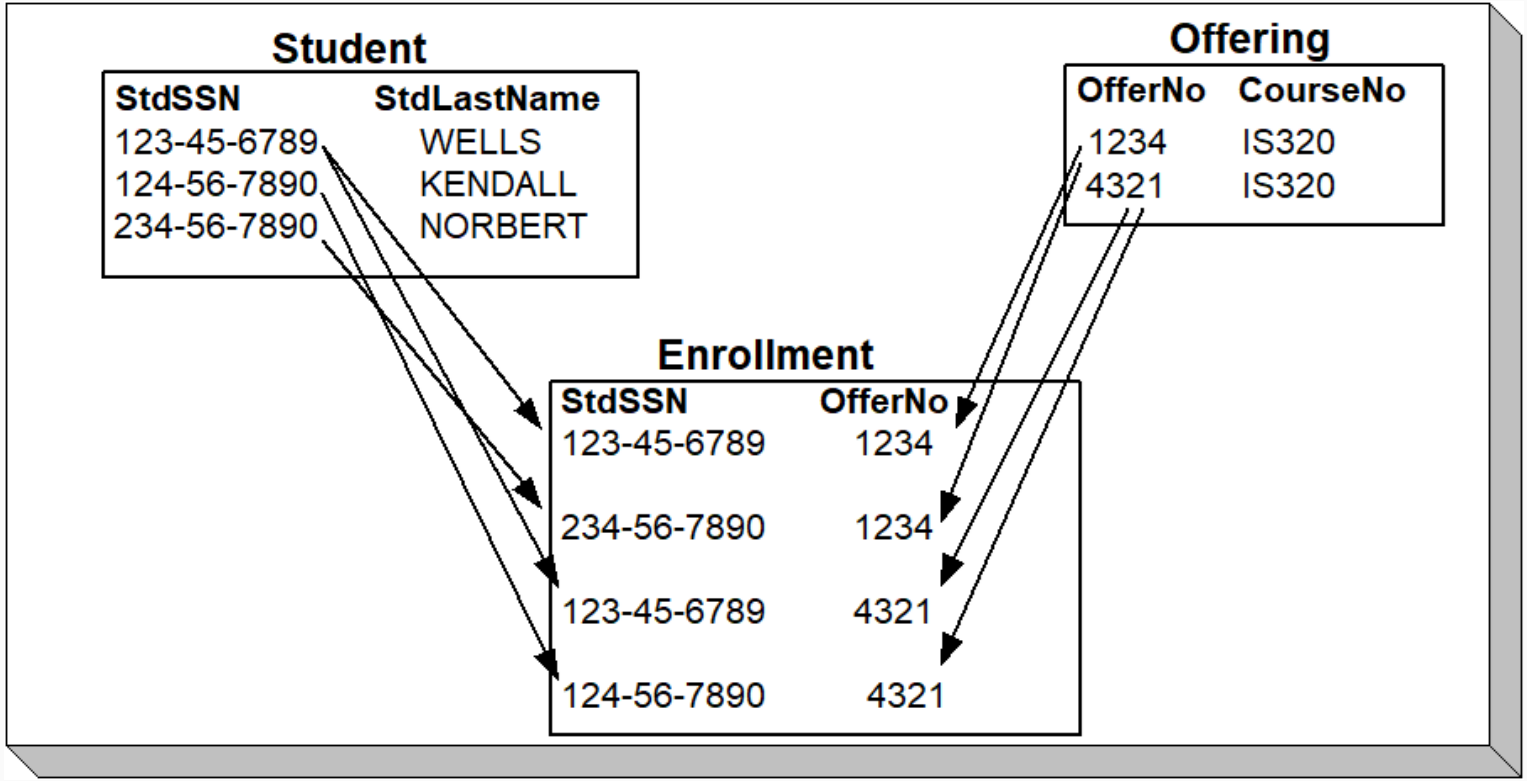
Create Table Statement

```
CREATE TABLE Student
(     StdSSN          CHAR(11),
      StdFirstName    VARCHAR(50),
      StdLastName     VARCHAR(50),
        StdCity                 VARCHAR(50),
      StdState        CHAR(2),
        StdZip        CHAR(10),
        StdMajor          CHAR(6),
      StdClass    CHAR(6),
        StdGPA                  DECIMAL(3,2)    )
```

Common Data Types

- CHAR(L)
- VARCHAR(L)
- INTEGER
- FLOAT(P)
- Date/Time: DATE, TIME, TIMESTAMP
- DECIMAL(W, R)
- BOOLEAN

Relationships



Alternative Terminology

Table-oriented	Set-oriented	Record-oriented
Table	Relation	Record-type, file
Row	Tuple	Record
Column	Attribute	Field

Integrity Rules

- Entity integrity: primary keys
  - Each table has column(s) with unique values
  - Ensures entities are traceable
- Referential integrity: foreign keys
  - Values of a column in one table match values from a source table
  - Ensures valid references among tables

Formal Definitions I

- Superkey: column(s) with unique values
- Candidate key: minimal superkey
- Null value: special value meaning value unknown or inapplicable
- Primary key: a designated candidate key; cannot contain null values
- Foreign key: column(s) whose values must match the values in a candidate key of another table

primary key candidate key

Student

ID	SSN	Name	Address	Phone	GPA
12345	111-11..	John D	123 My Ln	123-4567	3.5
56789	123-21...	Jane D	321 YourWay	876-5678	3.8

primary key candidate key

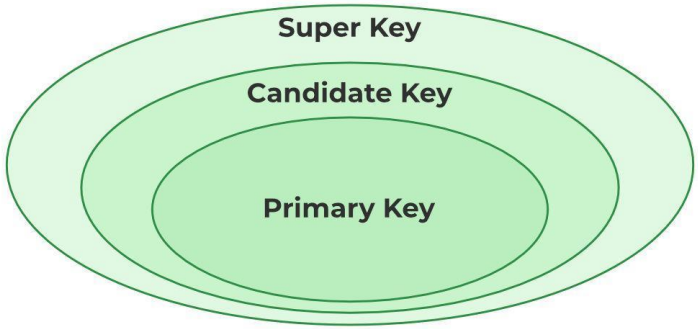
Course

CID	CourseNo	Name	Hours
2342	CS457	Database	4
1235	CS255	Comp Arch I	4

foreign key key

GradeReport

ID	CID	Semester	Grade
12345	2342	Fall 02	W
12345	2342	Fall 03	B+



Formal Definitions II

- Entity integrity
  - No two rows with the same primary key value
  - No null values in a primary key
- Referential integrity
  - Foreign keys must match candidate key of source table
  - Foreign keys in some cases can be null

ASSIGNMENT

1. MySQL already installed
2. Learn and prepare syntax of SQL (DDL & DML)
   CREATE, ALTER, DROP, TRUNCATE
   INSERT, SELECT, UPDATE, DELETE

Student will share their screen to prove this assignment